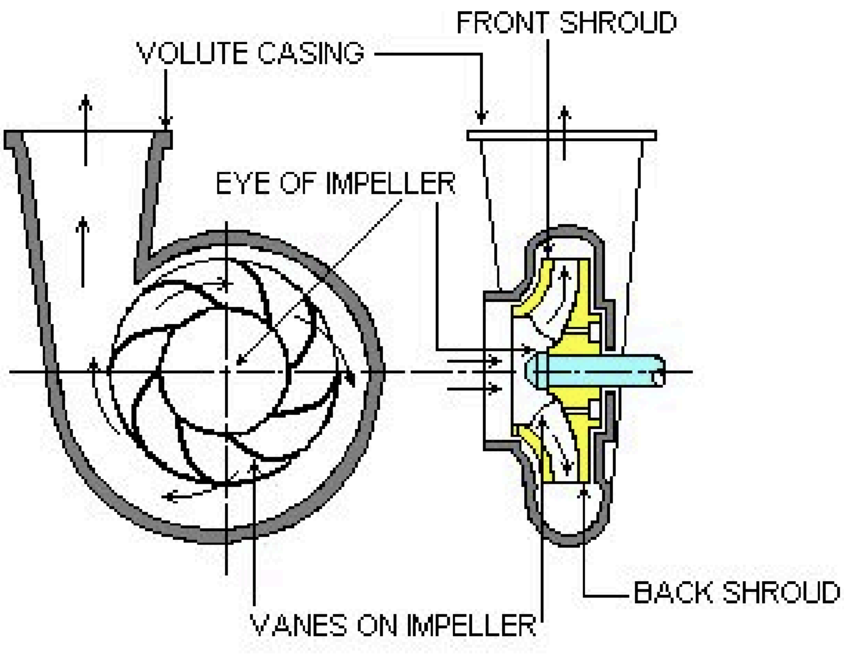
Centrifugal Pumps

Centrifugal pumps are another kind of fluid pump different to positive displacement pumps, known as rotodynamic pumps. From the name, they use rotational motion to transfer energy into the fluid.

There are three kinds of rotodynamic pumps, centrifugal/radial flow pumps, axial flow pumps and mixed flow, or propeller pumps. These kinds of pumps are typically used in water treatment, irrigation, chemical engineering and cooling systems.

The most common type of rotodynamic pumps is the centrifugal pump, typically running at high speeds of 2900 rpm for smaller pumps, to 1450rpm and 960 rpm for larger pumps.

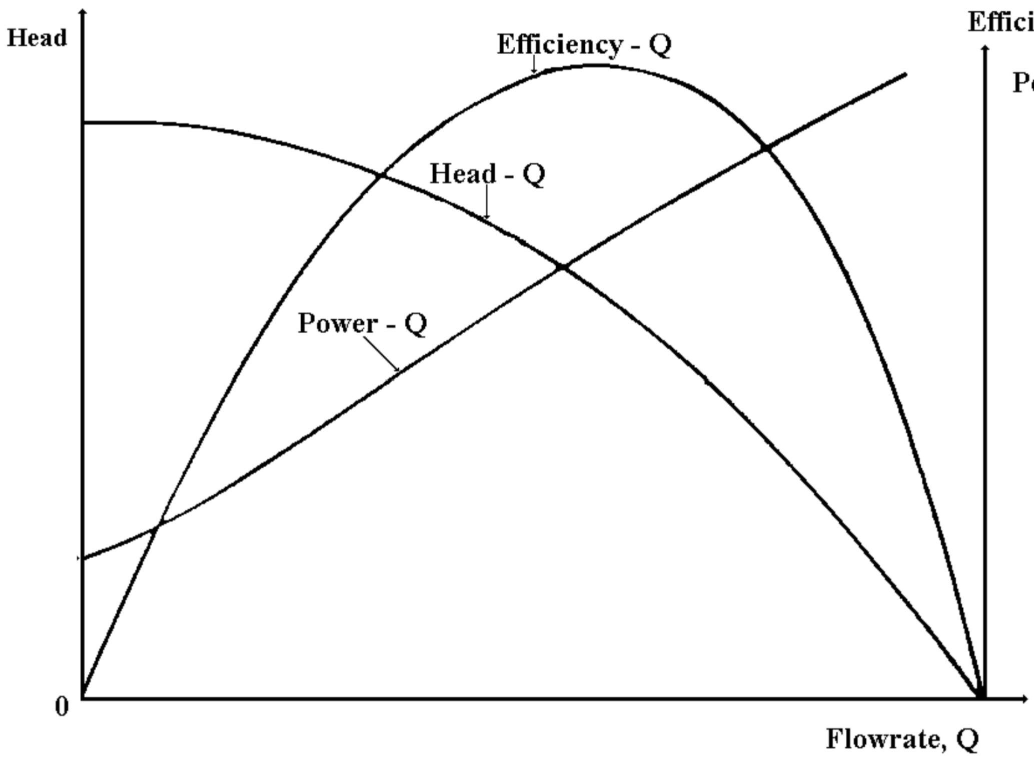


Centrifugal pumps allow fluid to flow into the center, or the eye of the impeller. Due to the high speed of the impeller, fluid flows radially outward, causing an increase in fluid velocity and pressure. The increased kinetic energy in the fluid is also partially converted into pressure energy, as the fluid contacts the spinning vanes on the impeller and due to the volute. The resulting output is high pressure fluid, but low flow rates.

For axial rotodynamic pumps, energy is transferred to the fluid by the shape of the vanes as the fluid deflects on them, and is output through the axis of the impeller. These pumps tend to produced high flow rates, but lower pressure increases.

Mixed flow rotodynamic pumps transfer energy to the fluid by both centrifugal force and the deflection of the fluid on the vanes. Thus, the resulting flow rate and pressure produced sits in the middle of centrifugal pumps and axial rotodynamic pumps.

The relationship between the pressure and flow rate can be determined by the H-Q curve, which can be determined experimentally or are given by the pump manufacturer. Another curve that can also be determined is the efficiency-flow rate curve, or $\eta - Q$ curve. Both curves talk about the performance of the pump.



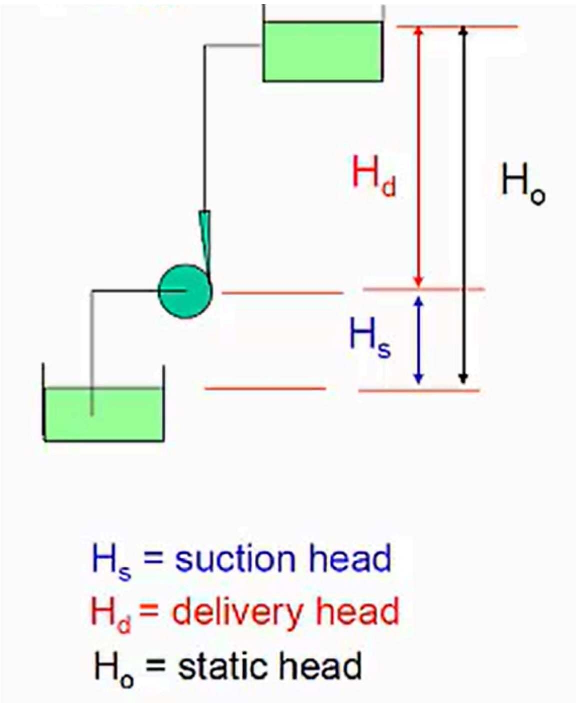
Typically, the H-Q curve has the pressure being negatively related to the flowrate, with the maximum pressure happening when there is no flow through the pump. This pressure is also known as the shut-off head. Note that this pressure is more of the maximum pressure the pump can produce at different flowrates, rather than the pressure being produced at that flowrate. Thus, the head is more dependent on the pump's properties.

Furthermore, a reason for this negative relationship for the H-Q curve is because of energy conservation, since $H \times Q = N/m^2 \times m^3/s = Nm/s = W$, which is a quantity of power, that has to be conserved.

For the efficiency curve, there is typically a maximum point where a certain flow rate gives the highest efficiency. From this flowrate, the point on the H-Q curve of the same flowrate is the best efficiency point, BEP.

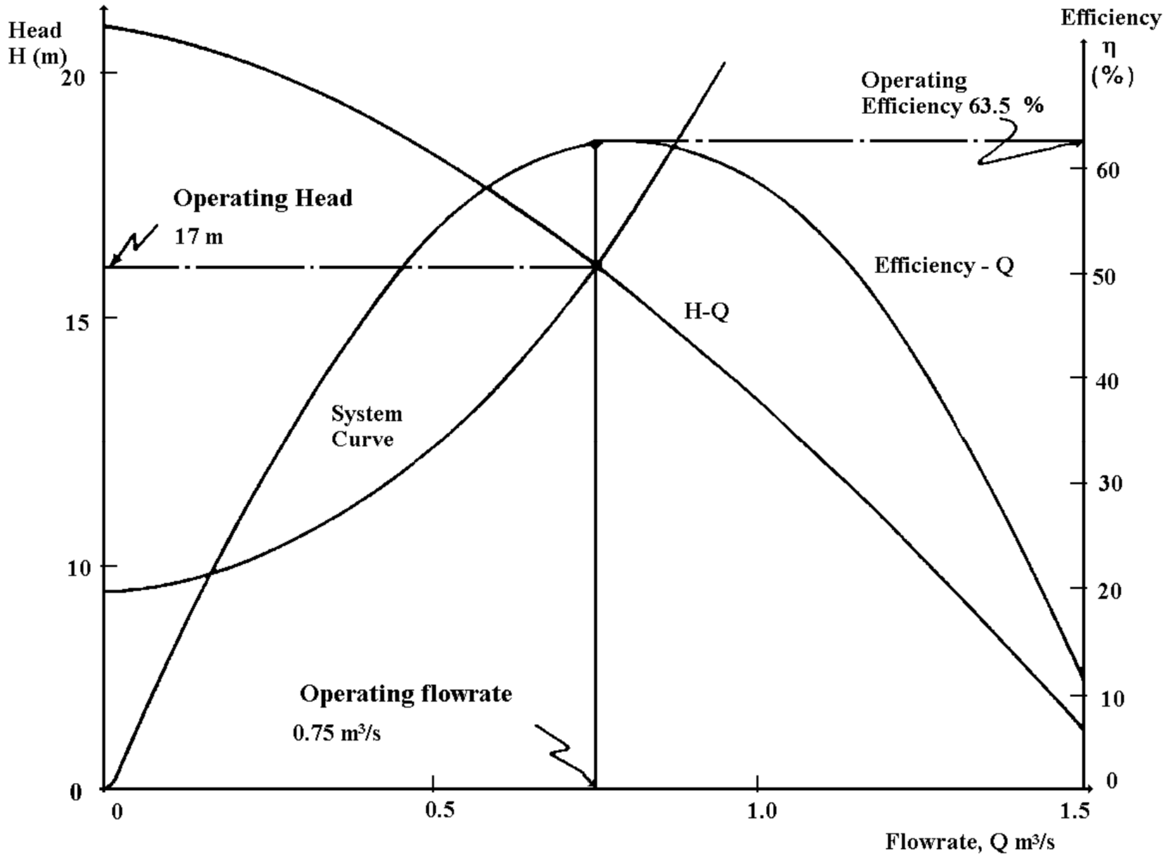
To move this fluid through a system via the pump, energy in the form of pressure head is imparted into the fluid at different flow rates. This energy comprises of two components.

Firstly, the static head, which is the output measured pressure by the pump, either as the vertical distance of the delivery level or the height of the end of the pipe delivery system.



The static head can also be thought of as the net vertical distance the pump needs to overcome.

Secondly, the dynamic head, more specifically the frictional losses which are proportional to the velocity of the flow. This can also be thought of as the pump needing to provide enough pressure to overcome the losses. Both of these heads combine to make the total head, which can be plotted on the same axes earlier, known as the system curve.



Notice how the system curve increases with flowrate, as a higher flow rate causes higher speeds, which increases the dynamic head and thus the total head. This is different to the H-Q curve, which indicates maximum head, while the system curve determines current working head.

Thus, there will be some intersection point between the two curves. The intersection point of the two curves is the operating point, which gives the operating flowrate. Hence, the operating efficiency can be found by using the operating flowrate.

Since the system curve is typically dependent on the system the pump is being used in (piping layout, etc.), while the H-Q curve is dependent on the pump's properties, the operating point may not necessarily be suitable for the system (different pump or different system may need to be used to change the curves). To determine the suitability of the obtained operating point, the pump selected needs to be compatible with the fluid used, the operating flow rate should be close to or ideally higher than the required flow rate, is close to the BEP, and does not experience cavitation.

Using the curves, the input or output power can be determined at any point/flow rate. The output power can be found as

$$P_{\text{out}} = \rho g H Q$$

where H and Q can be found graphically. To find the input power,

$$P_{\text{in}} = \frac{P_{\text{out}}}{\eta}$$

where η can be determined graphically at the value of Q used.

However, even when a specific pump has found to be suitable, the required flow rate may change later in operation. Thus, the flowrate can be changed with different methods, depending on the required changes, which may be short-term/permanent, small/substantial and increase/decrease. The pressure change due to changes in flow rate, which have to be considered for the flow rate change methods.

Throttling the discharge valve involves adjusting the output valve slightly to cause the flow rate to decrease slightly. It is one of the simplest methods to adjust the flow rate, however, is only suitable for small changes in flow rate. However, restricting the output valve's flow rate causes increased frictional losses, making the minimum energy required to pump the fluid to increase, hence why it is only suitable for small changes in flow rate.

Another method is by changing the speed of the pump, N. For a decrease in N, due to the speed of the fluid being roughly proportional to the pump's speed, and that the Darcy equation has head proportional to v^2, the head rapidly drops. Thus, N can only be decreased by small amounts. Similarly, decreases in N causes decreases in Q, as said earlier. Thus,

$$Q \propto N$$
$$H \propto N^2$$

The last method is changing the impeller's output diameter, by picking the version of a certain pump model with the needed diameter. For reductions in D, the head and flow rate drop proportionally to the square of the diameter, or the area of the output diameter. More specifically,

$$Q \propto D_2^2$$
$$H \propto D_2^2 \text{ for small values of } D_2$$

Pumps may also be used in conjunction with other pumps, in series or parallel.

Parallel pumps are used to increase the flowrate of 50 to 75%; since parallel pipes have summative flowrates. Hence, by using the same method to sum flow rates via adding up horizontal lines, the new H-Q can be found. This method also assumes that further friction due to new fittings to join both pumps is negligible, and thus the intersection of the new H-Q curve and the system curve gives the new operating point.

Series pumps are found similarly, by finding the new H-Q curve by summing vertical distances, and then finding the new intersection point. This also assumes that the system curve stays the same.

Before using a pump, it needs to be primed, which is the process of filling the pump casing and the suction pipe with the working fluid. If this was not done, the pump would be rotating in air, which has too low of a density to develop the required pressure. Since these pumps are also designed to operate at the rated pressure, if the pressure is too low, there could be damage from the lack of cooling or lubrication from the lack of fluid.

Hence, after priming the pump, a foot valve and strainer is typically fitted onto it to ensure the fluid stays in the pump to keep it primed. Priming can either be done manually, with a by-pass line back into the pump, or an ejector.

One phenomena of centrifugal pumps is cavitation, where fluid at lower pressures starts to boil as its temperature reaches above the vapour point, causing bubbles of gas to form. However, as fluid moves to higher pressure zones, the bubbles collapse and form pressure waves, which causes vibrations and affects the efficiency of the pump, or damage.

One way to evaluate if cavitation is likely to occur is through the net positive suction head available, or $NPSH_{\text{av}}$.

Since cavitation occurs at the lowest point of pressure in the system, it usually occurs at the inlet (lowest point). Thus, at the inlet, the head is

$$H_{\text{inlet}} = H_{\text{atm}} - H_{\text{suction}} - H_f - H_v$$

To prevent cavitation, there needs to be some “clearance” between the inlet head and the vapour pressure head, for $H_{\text{inlet}} > H_{\text{vapour}}$. Thus,

$$H_{\text{reserve}} = H_{\text{inlet}} - H_{\text{vapour}}$$

This reserve/clearance head is the net positive suction head available, and thus

$$NPSH_{\mathrm{av}} = H_{\mathrm{atm}} - H_{\mathrm{suction}} - H_f - H_v - H_{\mathrm{vapour}}$$

The NPSH available is dependent on the setup of the pump system, like piping and suction head. However, the pump may not be able to supply enough excess pressure to prevent cavitation, as it has its own NPSH values, called $NPSH_{\mathrm{req}}$. Hence, to prevent cavitation,

$$NPSH_{\mathrm{av}} > NPSH_{\mathrm{req}}$$

Required NPSH is usually given as an NPSH-Q graph, from the manufacturer.

If cavitation is predicted, either $NPSH_{\mathrm{av}}$ can be increased or $NPSH_{\mathrm{req}}$ can be decreased.

$NPSH_{\mathrm{req}}$ is mainly reduced by using an inducer, but $NPSH_{\mathrm{av}}$ can be increased by changing any of the different portions of the system, by

reducing suction head by installing the pump as close to the suction level

reducing friction head/velocity head by reducing velocity, or increasing suction pipe diameter